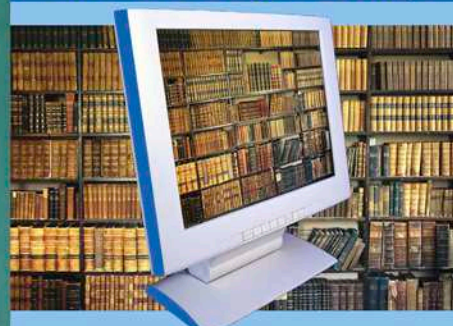


SKF

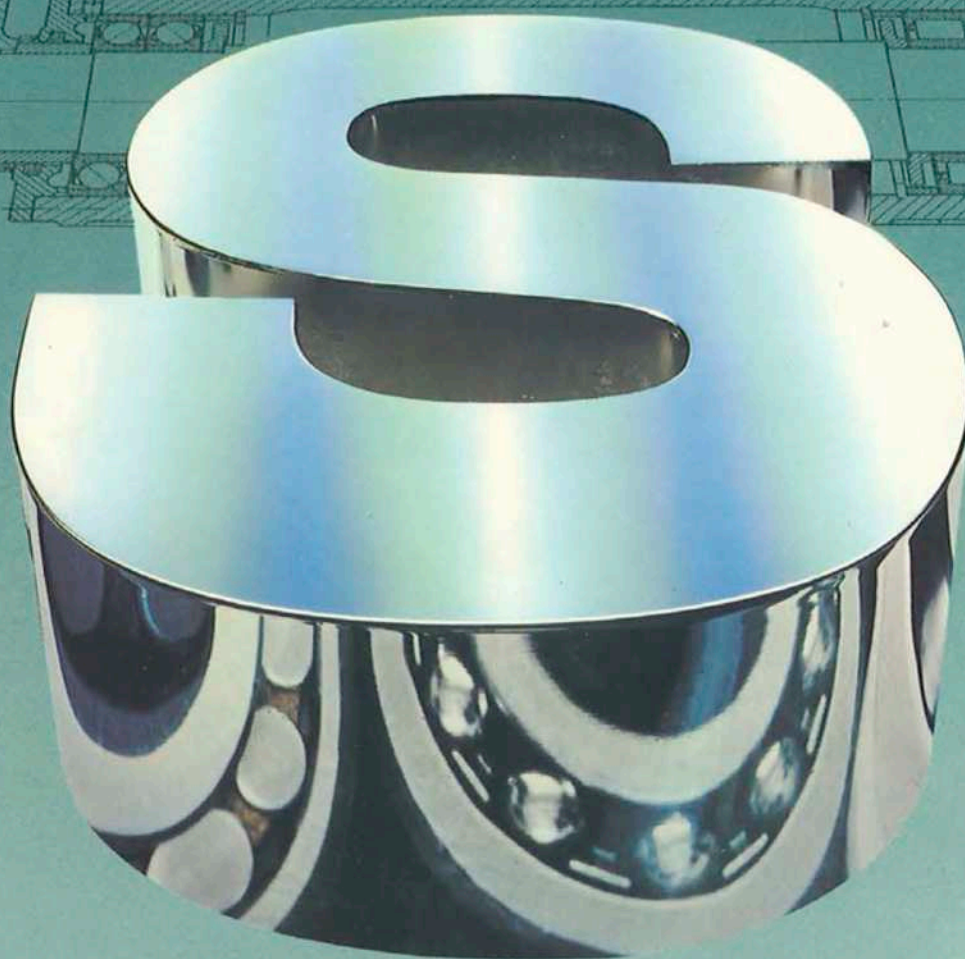
S-Range

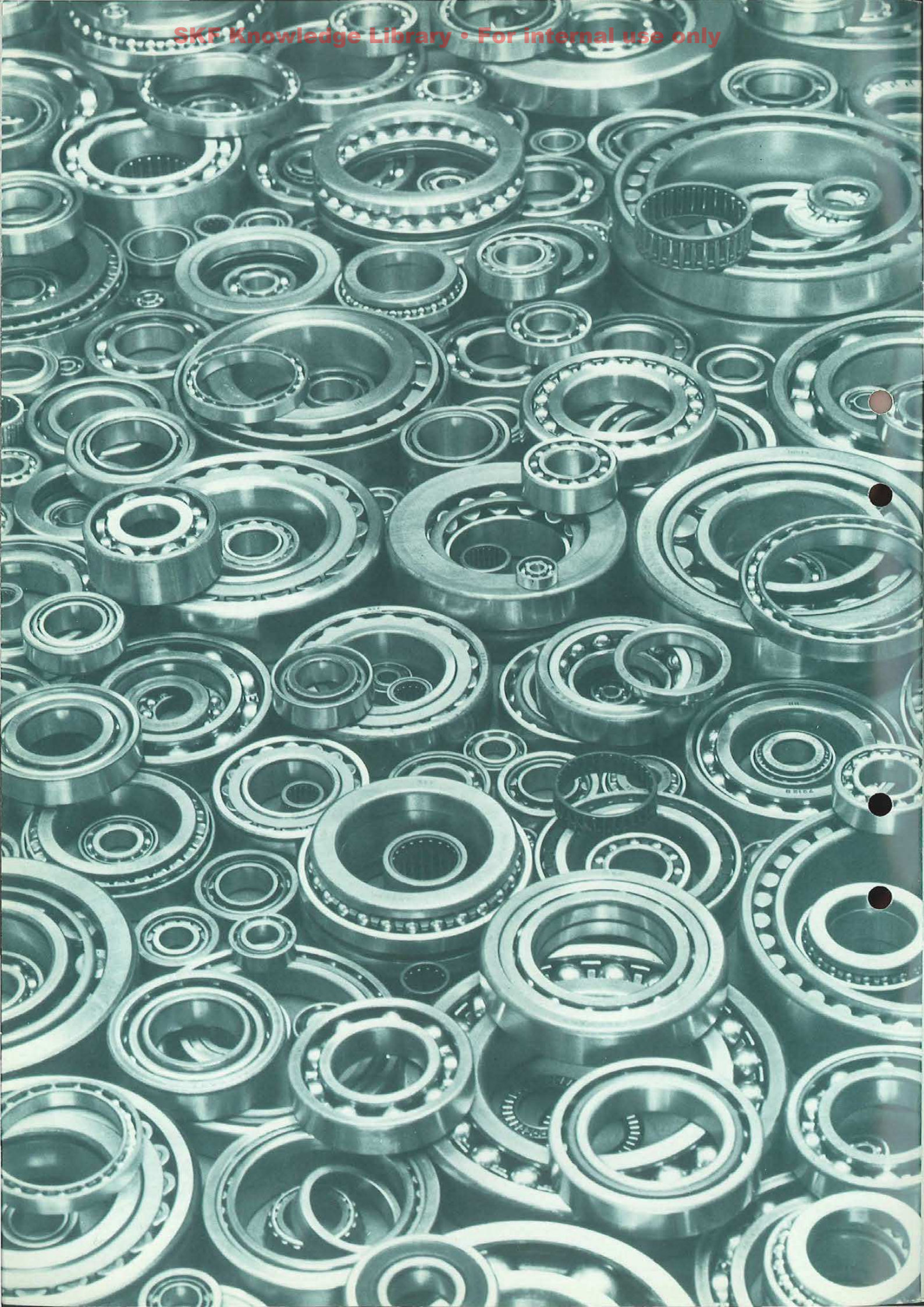
Bearings for fans

SKF Knowledge Library



For internal use only





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S-Range

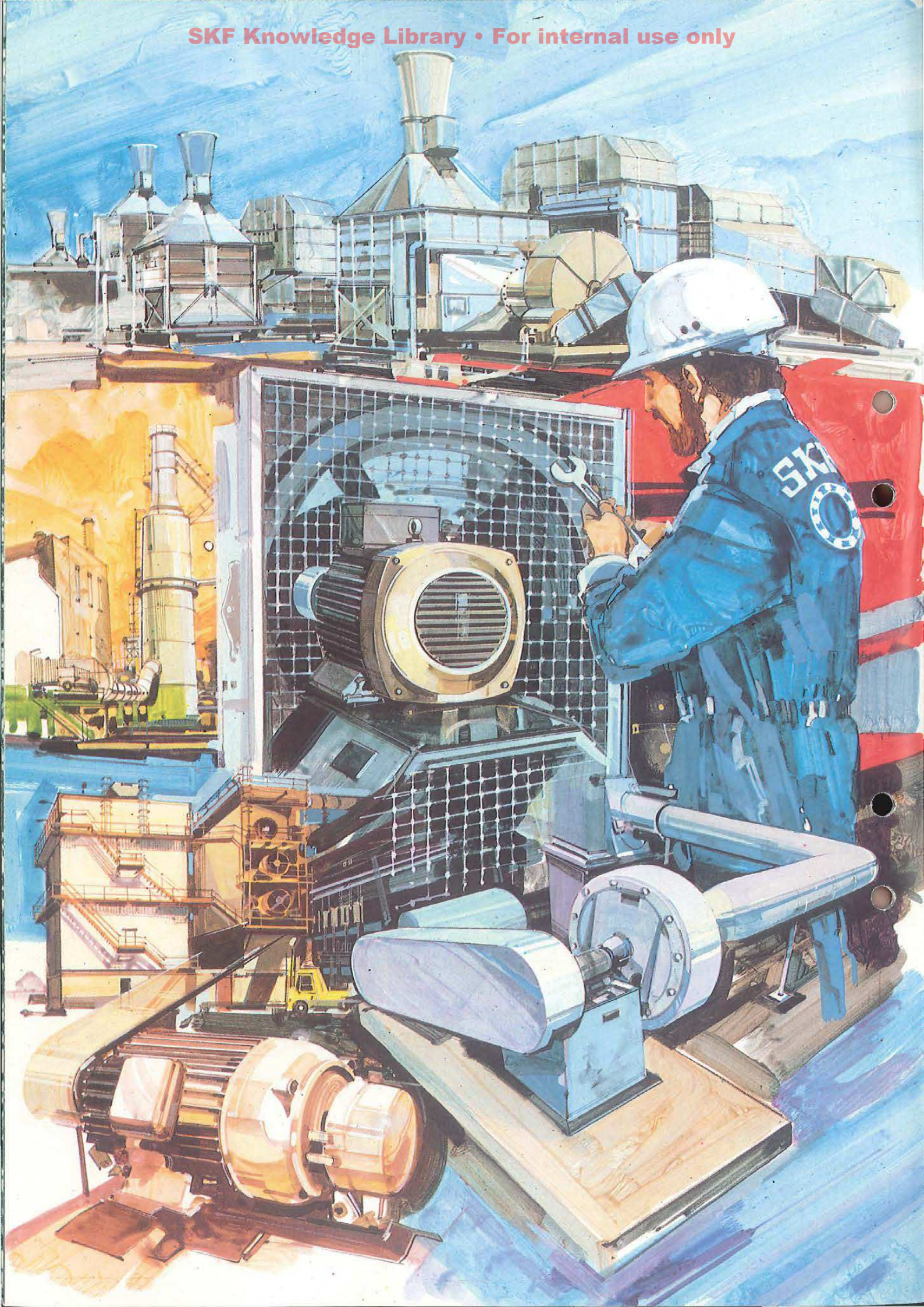
– a new concept

To further streamline SKF customer service a number of items from the broad assortment of SKF bearings has been carefully selected. These make up the S-Range, application orientated bearing ranges for specific application areas.

The selection of each S-Range item is based on wide SKF experience in the bearing application field, built through contact with bearing users all over the world.

It is SKF policy to supply virtually every type of bearing to meet customer requirements. But with S-Range, SKF now offers customers additional benefits:

- Simplified bearing selection and application design work coupled with confidence in the suitability of the range offered.
- Short delivery times.
- Long term supply stability.
- Worldwide availability.
- No minimum order quantities.
- Simplified ordering and stocking.



Fans – general

The large variety of fans at present available can be classified in different ways.

Based on the direction of the air flow or of the gas to be moved there are chiefly two types, namely centrifugal fans and axial flow fans. In more recent times tangential flow fans have also gained some importance; here the medium to be moved is introduced radially, transported over a certain range of the circumference by the blades of the impeller and then expelled radially.

The range of applications for fans is very wide and the demands placed on the bearings are equally varied, especially with regard to running noise and maintenance.

The impeller of a small fan is generally mounted on the extended shaft of an electric motor, whereas medium and large fans are fitted with separate bearing arrangements and are driven via couplings or V-belts. One field of application for medium fans is the ventilation of large rooms or workshops often in conjunction with air-conditioning plants. Large industrial fans are often classified according to their use, such as fresh air and hot gas fans for heating/power stations, cooling tower fans, converter fans, mine fans, tunnel fans etc.

The load acting on fan bearings is generally not too heavy. When selecting the bearings, however, the expected axial loads have to be considered for the majority of fans. It is also important to consider the extension of the shaft and temperature differences between the bearing inner and outer rings when designing the bearing arrangement.

It is recommended that bearings with C3 radial internal clearance are used when speeds exceed more than about 75 % of the limiting speed values given in the SKF General Catalogue; the same is true, at all speeds, for hot gas fans.

Generally the bearings can be used at operating temperatures up to +125 °C. Bearings with moulded cages of glass fibre reinforced polyamide can be used in the range -40 to +120 °C; higher temperatures, e.g. +130 or +140 °C, can be tolerated provided

they act only for short periods and that the intervening periods at lower temperatures are relatively long. For Y-bearings, the appropriate temperature range is given on page 11.

As the requisite basic rating life values for the bearings vary according to the desired duration of service and operational safety, the guideline values shown in the table below may be taken.

Type of operation	L_{10h} operating hours
Intermittent	$\leq 20\,000$
12 to 16 hours per day	40 000
24 hours per day with high operational reliability	$\geq 60\,000$

Calculation methods

It is not always easy to find the exact loads for the bearing calculation. However, the following simplified procedure is generally used.

1. Take the total mass of the fan wheel, the shaft and the coupling or pulley.
2. Calculate the additional forces generated by the belt pull or the coupling.

a) Belt drive

To calculate the belt pull first obtain the tangential load using the equation

$$F_t = 955 \times 10^4 \frac{P}{n r}$$

where

F_t = tangential load, N

P = power, kW

n = speed, r/min

r = pitch radius of pulley, mm

Then multiply the tangential load by a factor which varies with the type of drive used, i.e.

V-belt drive 2 to 2,5

Plain belt drive 3 to 4

b) Coupling drive

Calculate the radial reaction, which may result from misalignment or from the use of a semi-flexible coupling, using the empirical equation

$$F = 5\,600 \sqrt{\frac{P}{n}}$$

where

F = load applied perpendicularly to the coupling, N

P = power, kW

n = speed, r/min

3. Unbalance force

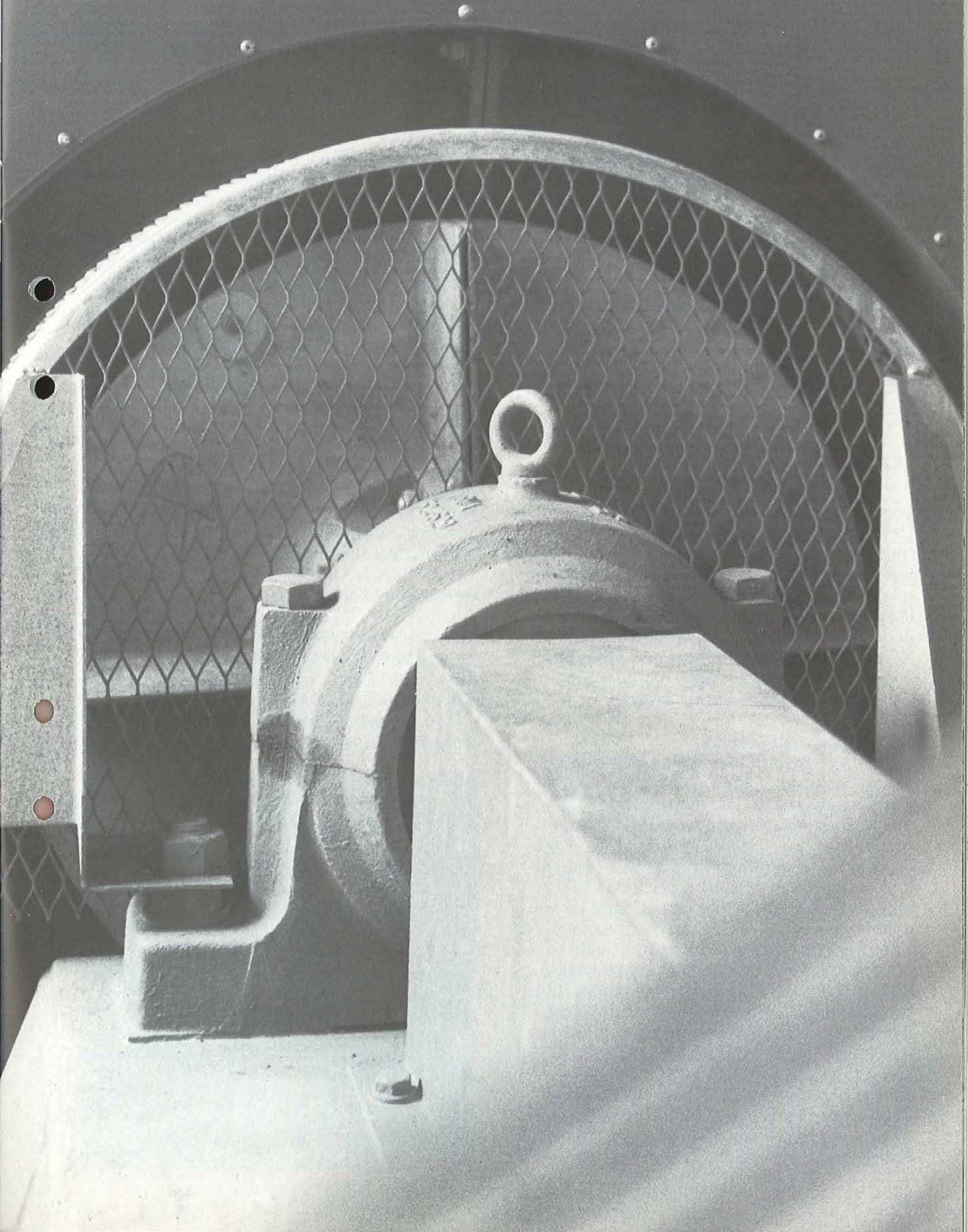
For a rough calculation of the unbalance force, expressed as a percentage of the mass of the fan wheel, the values shown in the adjacent table may be used.

4. Axial load

One of the bearings will also have to carry the thrust load which is generated when the fan is operating. When the fan axis is vertical the mass of the rotating parts, mentioned under 1, no longer acts as a radial force and has to be added to the thrust load of the operating fan.

The different loads have to be calculated for each bearing. The unbalance force, being a rotating load, should not be added directly to the loads of constant direction. To obtain the mean load for each bearing, therefore, a reduction factor f_m is needed, see pages 40 and 41 of the SKF General Catalogue.

Speed	Unbalance
r/min	%
1 000	30
2 000	60
3 000	90
4 000	125
5 000	150



S-Range of bearings for fans

The selection of bearings for a particular application is generally governed by the prevailing operating conditions, but may also be determined by various standards relating to that application. These factors, plus the wide experience gained by SKF over many years, have all been taken into account when selecting this S-Range. The complete S-Range of bearings for fans is shown, in matrix form, on pages 13 to 19.

A wide range of housings is available for the various bearing types, see pages 30 to 33.

Bearing types

Deep groove ball bearings

Deep groove ball bearings are very suitable for high speed applications; compared with self-aligning ball bearings they can also take relatively heavy axial loads. Unlike Y-bearings and self-aligning ball bearings, however, deep groove ball bearings have a limited ability to accommodate errors of alignment, and this must be taken into consideration when designing the bearing housing.

Self-aligning ball bearings

These bearings are included in this S-Range mainly because of their ability to accommodate misalignment. They are used for light and medium duty fans where high speeds predominate. A gradual changeover to cages of glass fibre reinforced polyamide (designation suffix TN9) is being carried out. Bearings with this cage which are already in production are shown in the matrix, see page 14.

Self-aligning ball bearings are usually mounted in SNA plummer block housings. Where the relubrication interval is short the housing may be fitted with a grease nipple during mounting. Excessive grease must be allowed to escape and SKF can supply a special design of SNA housing with grease escape valve, type TAV, see SKF General Catalogue. If housing type TC or TG is preferred, a grease escape hole should be cut or drilled in the housing.

Angular contact ball bearings

Angular contact ball bearings are suitable where moderate axial loads are involved. A pair of these bearings, mounted back-to-back, can be incorporated as locating bearings into two-bearing housings instead of deep groove ball bearings.

Cylindrical roller bearings

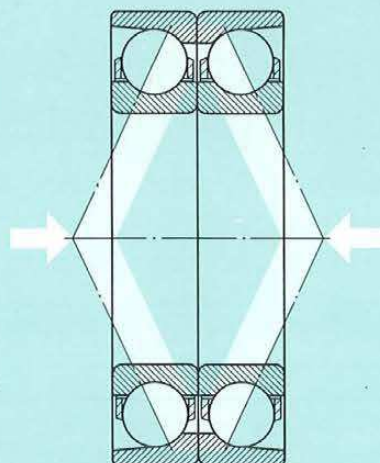
These bearings perform favourably under high speeds and loads. They are normally used as non-locating bearings in two-bearing housings where paired angular contact ball bearings or deep groove ball bearings are used as locating bearings.

SKF has recently introduced a new standard design of cylindrical roller bearing which has increased radial and axial load carrying capacities and a cage of glass fibre reinforced polyamide (ECP design). Those sizes which are already available are shown in the matrix on page 16.

Spherical roller bearings

Spherical roller bearings are used for larger fans where a high carrying capacity and misalignment accommodation are required.

The CC type spherical roller bearings included in this S-Range have a high speed



Back-to-back arrangement

capability and give low running temperatures even under moderate axial loads.

SNA plummer blocks, often in connection with a grease escape valve, type TAV, are usually recommended. For shaft diameters over 140 mm at moderate speeds, SD 31 housings which incorporate labyrinth seals (designation suffix TS) are available. Where speeds are moderate to high, SOFN housings are recommended.

Spherical roller thrust bearings

Spherical roller thrust bearings are used in horizontal and vertical fans where heavy axial loads have to be carried. The new E-type bearing has a higher load carrying capacity and also lower friction than the earlier designs.

Generally spherical roller thrust bearings are utilised to take axial loads only, whilst the radial loads are taken by deep groove ball bearings, cylindrical roller bearings or spherical roller bearings.

Y-bearings

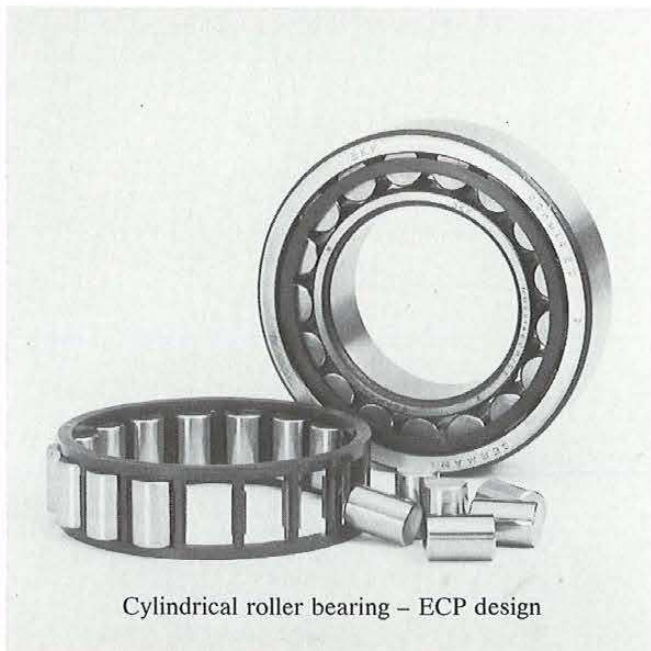
SKF Y-bearings are basically deep groove ball bearings which are fitted with very effective seals and are supplied already charged with the correct amount of grease. The permissible operating temperature range is -30 to $+110$ °C.

These bearings can accommodate a certain degree of initial misalignment by virtue of the sphered outside diameter and the corresponding sphered seating in the housing.

The limiting speeds for Y-bearings are governed by the method used to locate them on the shaft, see the SKF catalogue "Y-bearing units".

Bearing data

Data on the bearings included in this S-Range with the exception of Y-bearings will be found in the SKF General Catalogue. For data on Y-bearings reference should be made to the SKF catalogue "Y-bearing units".



Cylindrical roller bearing – ECP design

S-Range matrices

As far as possible the bearings included in the S-Range matrices are presented in the order in which they will be found in the SKF General Catalogue. They are shown in Dimension Series order (light section series to the left, heavy to the right), and bearings with Normal radial internal clearance are shown first, followed by those with clearance other than Normal.

M	Machined brass cage.
P	Cage of glass fibre reinforced polyamide.
TN9	Cage of glass fibre reinforced polyamide.
W33	Lubrication groove and three holes in the outer ring.

Bearing designations

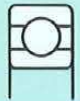
The matrix headings show bearing designations without the size figures. A lighter coloured square indicates the position for the size figures which are shown to the right of the matrices.

Designation suffixes

The following designation suffixes will be found in the matrices.

B	With single row angular contact ball bearing denotes a contact angle of 40°.
C3	Radial internal clearance greater than Normal.
E	With spherical roller thrust bearings denotes new internal design to give improved load carrying capacity with reduced friction.
EC	With cylindrical roller bearings denotes increased radial and axial load carrying capacities.
G	Single row angular contact ball bearings with ring faces adjusted so that any two bearings can be mounted in pairs, i.e. in tandem, back-to-back or face-to-face.
K	Tapered bore, taper 1:12 on diameter.

S-Range – Fans Deep groove ball bearings

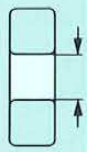


 d (mm)	Series			
	With Normal clearance		With C3 clearance	
	62	63	62/C3	63/C3
10				
12				
15				
17				
20				
25				
30				
35				
40				
45				
50				
55				
60				
65				
70				
75				
80				
85				
90				
95				
100				
110				
120				
130				
140				
150				
160				
170				
180				
190				
200				
220				
240				
260				
280				
300				
320				
340				
360				
380				
400				
420				
440				
460				
480				
500				
530				
560				
600				
630				
670				
710				
750				

000 XX

S-Range – Fans

Self-aligning ball bearings



Series
With tapered bore
and C3 clearance

d
(mm)

12 □ K/C3
12 □ KTN9/C3
22 □ K/C3
22 □ KTN9/C3
22 □ KM/C3
13 □ K/C3
23 □ K/C3
23 □ KM/C3

000 □ XX

10
12
15
17
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
110
120
130
140
150
160
170
180
190
200
220
240
260
280
300
320
340
360
380
400
420
440
460
480
500
530
560
600
630
670
710
750

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14
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16
17
18
19
20
22
24
26
28
30
32
34
36
38
40
44
48
52
56
60
64
68
72
76
80
84
88
92
96
/500
/530
/560
/600
/630
/670
/710
/750

S-Range – Fans

Angular contact ball bearings



d (mm)		Series																000 XX
		72 BG 72 BGM																
10																		00
12																		01
15																		02
17																		03
20																		04
25																		05
30																		06
35																		07
40																		08
45																		09
50																		10
55																		11
60																		12
65																		13
70																		14
75																		15
80																		16
85																		17
90																		18
95																		19
100																		20
110																		22
120																		24
130																		26
140																		28
150																		30
160																		32
170																		34
180																		36
190																		38
200																		40
220																		44
240																		48
260																		52
280																		56
300																		60
320																		64
340																		68
360																		72
380																		76
400																		80
420																		84
440																		88
460																		92
480																		96
500																		/500
530																		/530
560																		/560
600																		/600
630																		/630
670																		/670
710																		/710
750																		/750

S-Range – Fans Cylindrical roller bearings



 d (mm)	Series With C3 clearance		 0000 XX
	NU 22 □ ECP/C3 NU 22 □ /C3	NU 3 □ ECP/C3 NU 3 □ /C3	
10			00
12			01
15			02
17			03
20			04
25			05
30			06
35			07
40			08
45			09
50			10
55			11
60			12
65			13
70			14
75			15
80			16
85			17
90			18
95			19
100			20
110			22
120			24
130			26
140			28
150			30
160			32
170			34
180			36
190			38
200			40
220			44
240			48
260			52
280			56
300			60
320			64
340			68
360			72
380			76
400			80
420			84
440			88
460			92
480			96
500			/500
530			/530
560			/560
600			/600
630			/630
670			/670
710			/710
750			/750

The new generation of SKF standard cylindrical roller bearings, identified by the designation suffix EC, will gradually replace existing cylindrical roller bearings. Please check availability with SKF.

S-Range – Fans

Spherical roller bearings



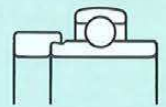
 d (mm)	Series		With tapered bore and Normal clearance		With tapered bore and C3 clearance		 000 XX
	With cylindrical bore and C3 clearance		With tapered bore and Normal clearance		With tapered bore and C3 clearance		
	222 □ CC/C3W33		222 □ CCK		231 □ CCK/C3W33		
	223 □ CC/C3W33		222 □ CCK/W33		231 □ CACK/C3W33		
			223 □ CCK		222 □ CCK/C3		
			223 □ CCK/W33		222 □ CCK/C3W33		
					223 □ CCK/C3		
					223 □ CCK/C3W33		
10							00
12							01
15							02
17							03
20							04
25							05
30							06
35							07
40							08
45							09
50							10
55							11
60							12
65							13
70							14
75							15
80							16
85							17
90							18
95							19
100							20
110							22
120							24
130							26
140							28
150							30
160							32
170							34
180							36
190							38
200							40
220							44
240							48
260							52
280							56
300							60
320							64
340							68
360							72
380							76
400							80
420							84
440							88
460							92
480							96
500							/500
530							/530
560							/560
600							/600
630							/630
670							/670
710							/710
750							/750

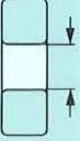
S-Range – Fans Spherical roller thrust bearings



		Series																000 XX
d (mm)	293□ E																	
10																	00	
12																	01	
15																	02	
17																	03	
20																	04	
25																	05	
30																	06	
35																	07	
40																	08	
45																	09	
50																	10	
55																	11	
60																	12	
65																	13	
70																	14	
75																	15	
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170																	34	
180																	36	
190																	38	
200																	40	
220																	44	
240																	48	
260																	52	
280																	56	
300																	60	
320																	64	
340																	68	
360																	72	
380																	76	
400																	80	
420																	84	
440																	88	
460																	92	
480																	96	
500																	/500	
530																	/530	
560																	/560	
600																	/600	
630																	/630	
670																	/670	
710																	/710	
750																	/750	

S-Range – Fans Y-bearings



 d (mm)	Series																			
	YEL 2 ☐																			
10																				00
12																				01
15																				02
17																				03
20																				04
25																				05
30																				06
35																				07
40																				08
45																				09
50																				10
55																				11
60																				12
65																				13
70																				14
75																				15
80																				16
85																				17
90																				18
95																				19
100																				20
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160																				32
170																				34
180																				36
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200																				40
220																				44
240																				48
260																				52
280																				56
300																				60
320																				64
340																				68
360																				72
380																				76
400																				80
420																				84
440																				88
460																				92
480																				96
500																				/500
530																				/530
560																				/560
600																				/600
630																				/630
670																				/670
710																				/710
750																				/750



Bearing arrangement design

Bearing arrangements for fans may be designed in a variety of ways depending upon the size, operating conditions etc. and a wide range of rolling bearings would be required to cover every type of application.

Some typical bearing arrangements, which include S-Range bearings and appropriate housings, are included in this leaflet.

Particulars of suitable bearing seating tolerances are given in the application notes but it should be remembered that the selection of tolerances relates, in these instances, to the sizes of bearings applied. Consequently, for similar applications using bearings of other sizes, it may be necessary to select different tolerances for the shaft seatings. Recommendations regarding tolerance selection are to be found in the SKF General Catalogue. However, it is advisable to consult SKF about tolerances for fans for special operating conditions, e.g. hot gas fans.

A guide to the lubrication of the bearings in the various fans is also given in the notes, but SKF should be consulted whenever further information is desired concerning lubrication.

Bearing arrangements using Y-bearing units

Bearing selection

SKF Y-bearings are recommended for light duty fans with shaft diameters up to and including 50 mm, see matrix on page 19.

These bearings have a zinc coated bore and an extended inner ring with eccentric locking collar. A list of appropriate housings will be found on page 30.

Bearing arrangement

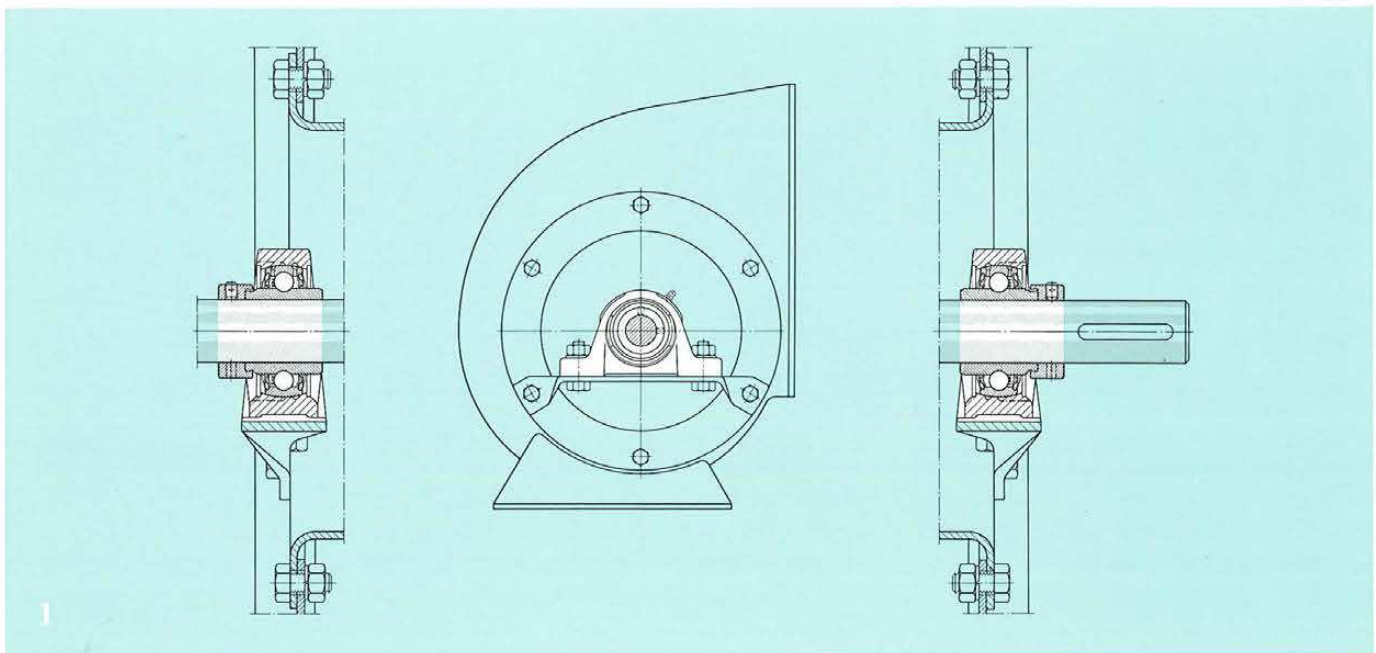
In the arrangement shown in fig 1 the fly-wheel of the fan is supported by Y-bearings mounted in cast iron housings. As both Y-bearings are located the sheet steel walls of the fan must accommodate possible thermal elongation of the shaft.

Tolerances

As the bearing bore tolerances are to plus limits to permit mounting on drawn steel shafts (say tolerance h9/IT5) a clearance fit is obtained. This leads to a slightly eccentric operation with resulting vibration, therefore the use of Y-bearings should be confined to low or medium speed operation.

Lubrication

Relubrication is not normally required as the bearings are supplied lubricated for life. However, if necessary, Y-bearings fitted in cast iron housings can be relubricated. Please see the SKF catalogue "Y-bearing units".



Bearing arrangements using SNA and SD plummer block housings

Bearing selection

Where silent running is stipulated with relatively high speeds, self-aligning ball bearings mounted on adapter sleeves are recommended for light and medium duty fans with shaft diameters up to and including 110 mm, see matrix on page 14. For heavy-duty fans spherical roller bearings mounted on adapter sleeves, are recommended, see matrix on page 17. A list of appropriate housings will be found on page 31.

Bearing arrangement

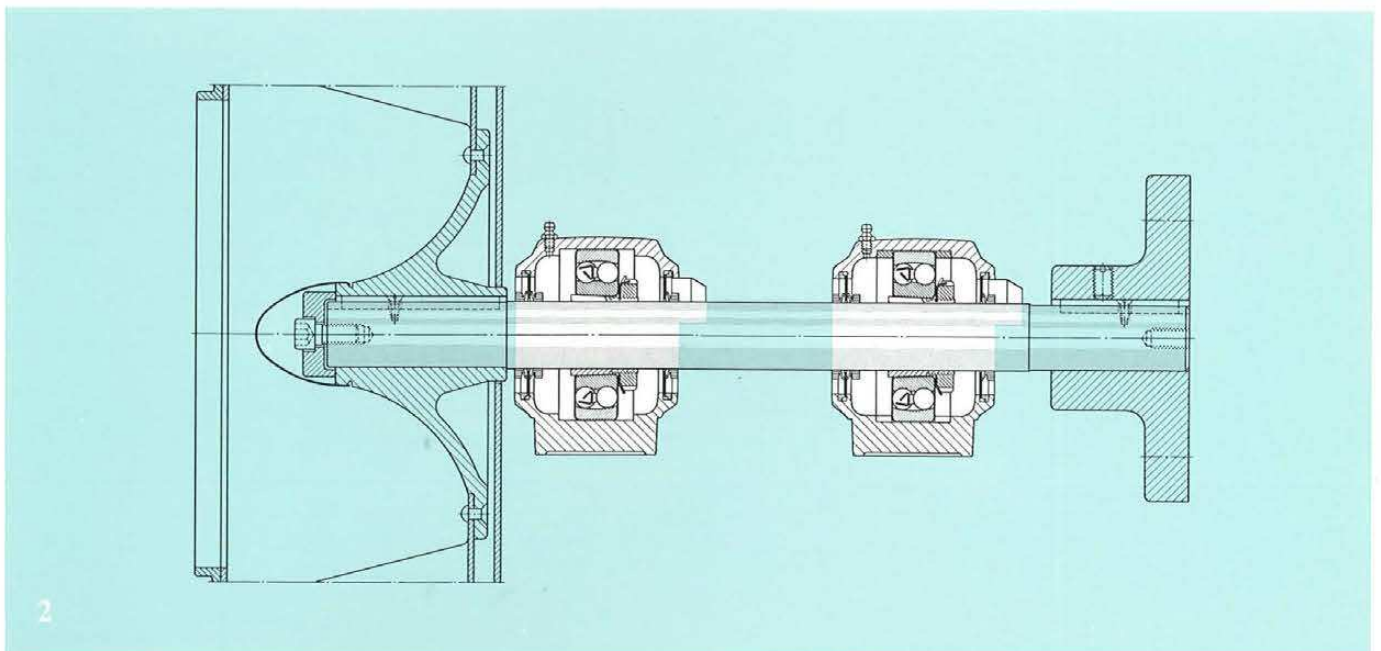
Normally the bearing is mounted in an SNA plummer block housing. The various types of seal are shown in the SKF General Catalogue. Relubrication can be arranged if there is a suitable grease escape arrangement for use with the various types of seal. Fig 2 shows an arrangement using a self-aligning ball bearing mounted in an SNA plummer block housing with grease escape valve, type TAV. The efficiency of relubrication has been much improved by mounting an extra V-ring inboard of the V-ring seal washer at the side where grease is supplied, so that grease can only leave the housing at the opposite side after passing through the bearing. It should be noted that grease is usually supplied to these housings on the side away from the lock nut.

Tolerances

Shaft		h9/IT5
Housing	Standard plummer block	H8
	Plummer block with grease escape valve	H7

Lubrication

A high quality lithium base grease, such as SKF LGMT2 or LGMT3, is normally recommended.



Bearing arrangements using two-bearing housings

Bearing selection

Deep groove ball bearings, paired angular contact ball bearings and cylindrical roller bearings, in different combinations, are all recommended for use in two-bearing housings with shaft diameters up to and including 120 mm for light to heavy duty fans. See matrices on pages 13, 15 and 16. Appropriate housings are listed on pages 32 and 33. It should be noted that SKF can also deliver complete units including shafts; please check availability with SKF before ordering.

Bearing arrangement

Fig 3 and 4 show arrangements which incorporate single row deep groove ball bearings. The bearings are mounted with axial clearance in the housings and up to and including sizes 6218 and 6316 they can be lightly preloaded by using springs in order to reduce noise. V-rings have been used at both sides in the illustrations, but the end covers for housings of series PDN 3 and PDR 3 are also provided with grooves so that felt rings can be used alternatively or additionally. The hot gas fan shown in fig 4 is protected on the outside by a heat-resistant wall; the shaft on the rotor side is drilled and filled with a similar heat-resistant material. A cooling ring, usually made of aluminium, has also been included between the impeller and bearing housing.

For heavy radial or axial loads, two-bearing housings can be fitted with paired angular contact ball bearings of series 72 as locating bearings and cylindrical roller bearings of series NU 22 or NU 22 ECP as non-

locating bearings. Fig 5 shows such a unit, series PDPF 22.

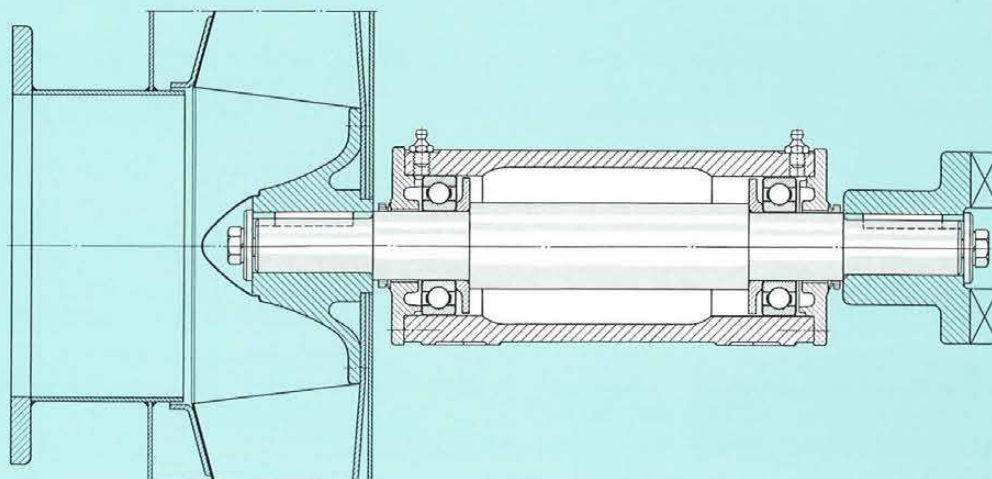
For heavier radial loads, the arrangement PDRJ 3, shown in fig 6, is more favourable. Here, cylindrical roller bearings of series NU 3 or NU 3 ECP are used as radial bearings, whilst a deep groove ball bearing of series 63 serves as the locating bearing.

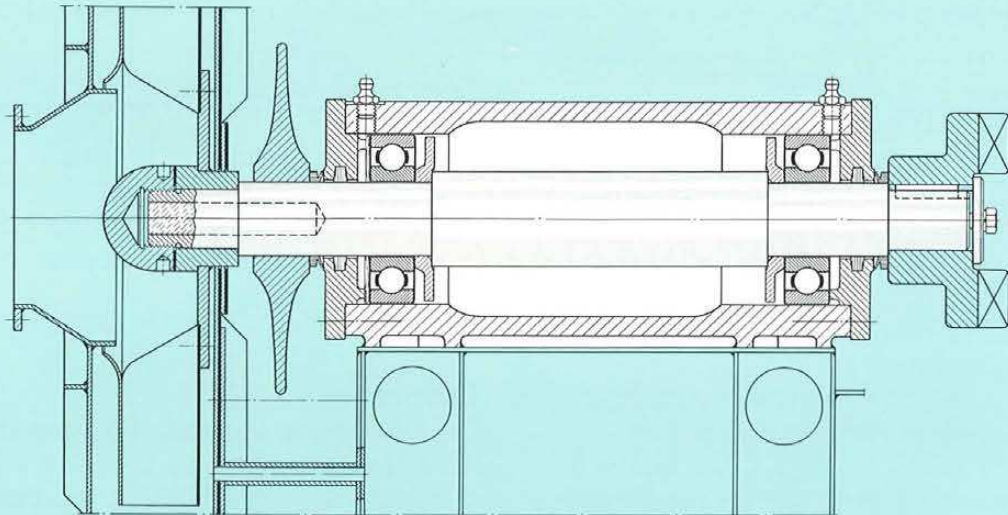
Tolerances

Shaft	k6
Housing	G6

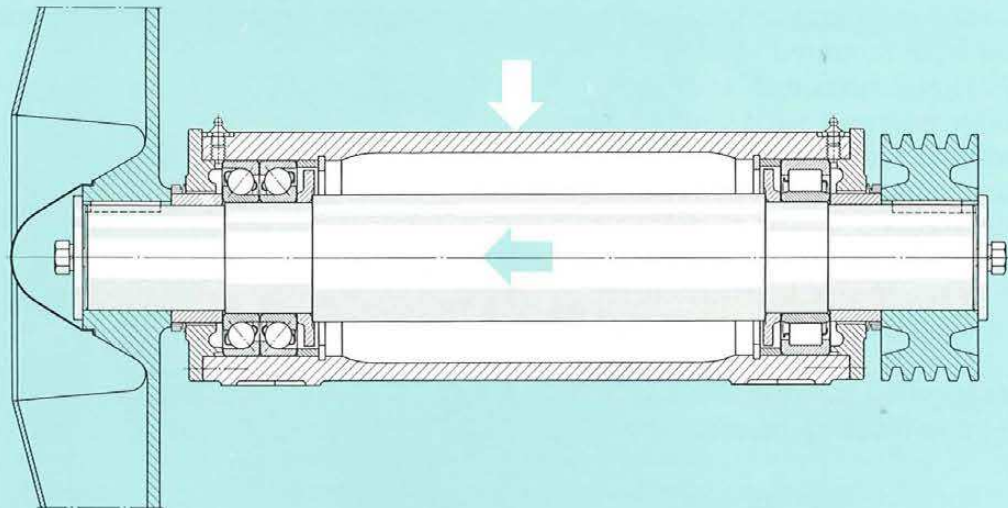
Lubrication

A high quality lithium base grease, such as SKF LGMT2 or LGMT3, is normally recommended.

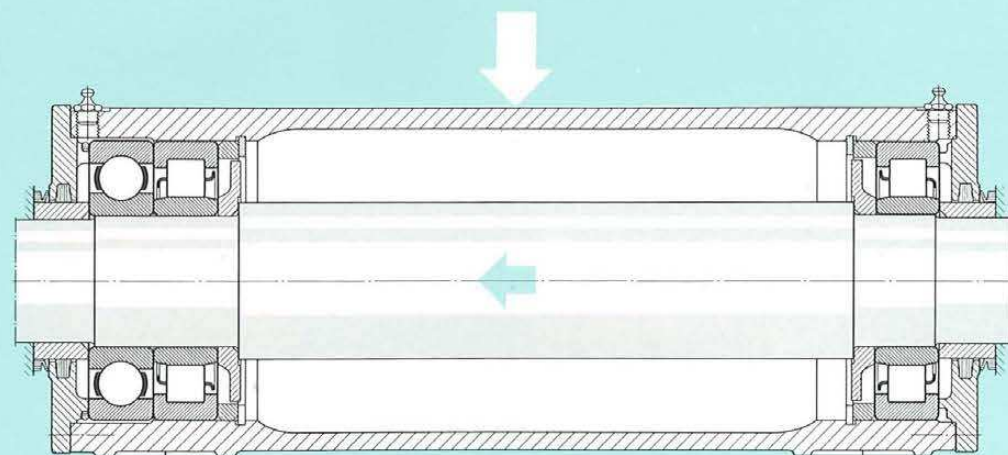




4



5



6

Bearing arrangements using SOFN plummer block housings

Bearing selection

Spherical roller bearings with cylindrical bore and also with tapered bore plus the relevant adapter sleeve, are recommended for the larger heavy-duty fans, see matrix on page 17. Appropriate housings will be found on page 33.

Bearing arrangement

Where long relubrication intervals are desirable oil lubrication is recommended and SOFN plummer block housings can be used. These have an adequate space for an oil reservoir and have been developed mainly for high speed fans. They are equipped with effective labyrinth seals to eliminate oil losses.

For applications where low vibration and silent operation are required, preference is given to the use of spherical roller bearings with cylindrical bore mounted in series SOFN 2 and SOFN 3 housings, see fig 7.

Spherical roller bearings with tapered bore mounted on adapter sleeves are frequently used where easy mounting is required. In this case series SOFN 5 housings should be chosen.

Each SOFN housing size is available in three slightly different designs:

Shaft end, non-locating bearing – suffix AL

Through shaft, non-locating bearing – suffix BL

Through shaft, locating bearing – suffix BF

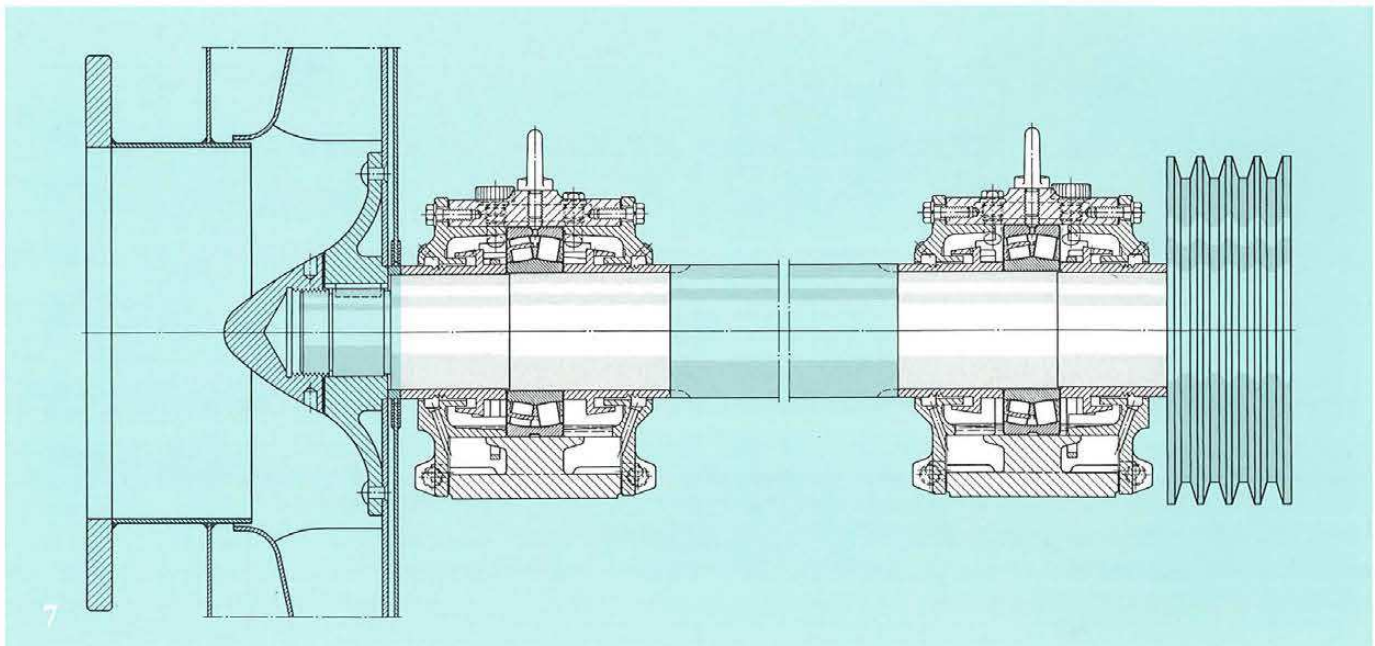
Tolerances

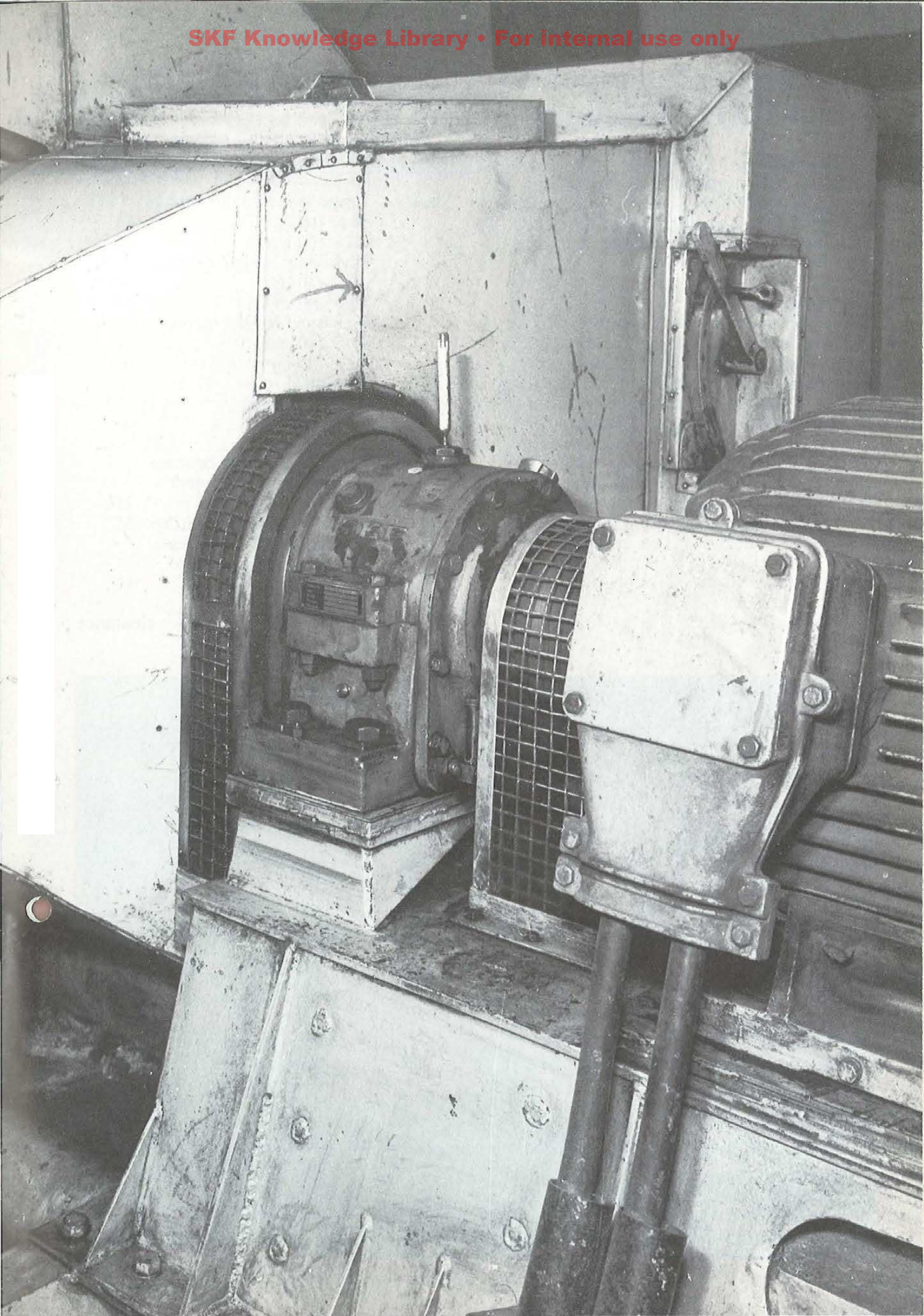
Shaft	Cylindrical seatings – direct mounting	m6
	Cylindrical seatings – mounting on sleeves	h9/IT5

Housing	F6
---------	----

Lubrication

Oil lubrication is used. To keep the bearing temperature as low as possible with the minimum amount of oil in the bearing, the oil is lifted from the reservoir to a collecting trough, as the shaft rotates, by a pick-up ring which hangs loosely on a sleeve on the shaft and dips into the oil in the lower half of the housing. The oil then passes through the bearing on its way back to the reservoir.





Bearing arrangements incorporating spherical roller thrust bearings

Bearing selection

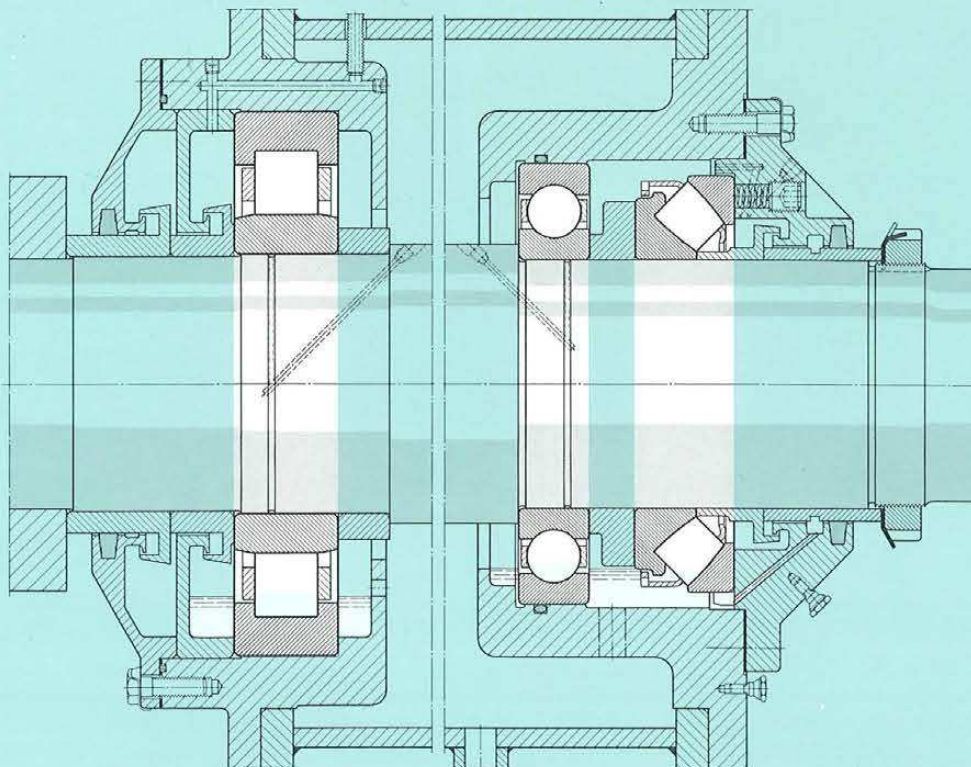
Spherical roller thrust bearings for use in conjunction with deep groove ball bearings, cylindrical roller bearings and spherical roller bearings, are shown in the matrix on page 18.

Bearing arrangement

When high axial forces have to be accommodated, it is sometimes necessary to use a thrust bearing for the support. Fig 8 and 9 show respectively a horizontal and a vertical fan, each fitted with a spherical roller thrust bearing. In each case, the spherical roller thrust bearing is radially free and therefore only axially loaded; the housing washer is loaded by using several springs, equally spaced around the periphery, to prevent the bearing from separating when the fan is started or the thrust load reversed.

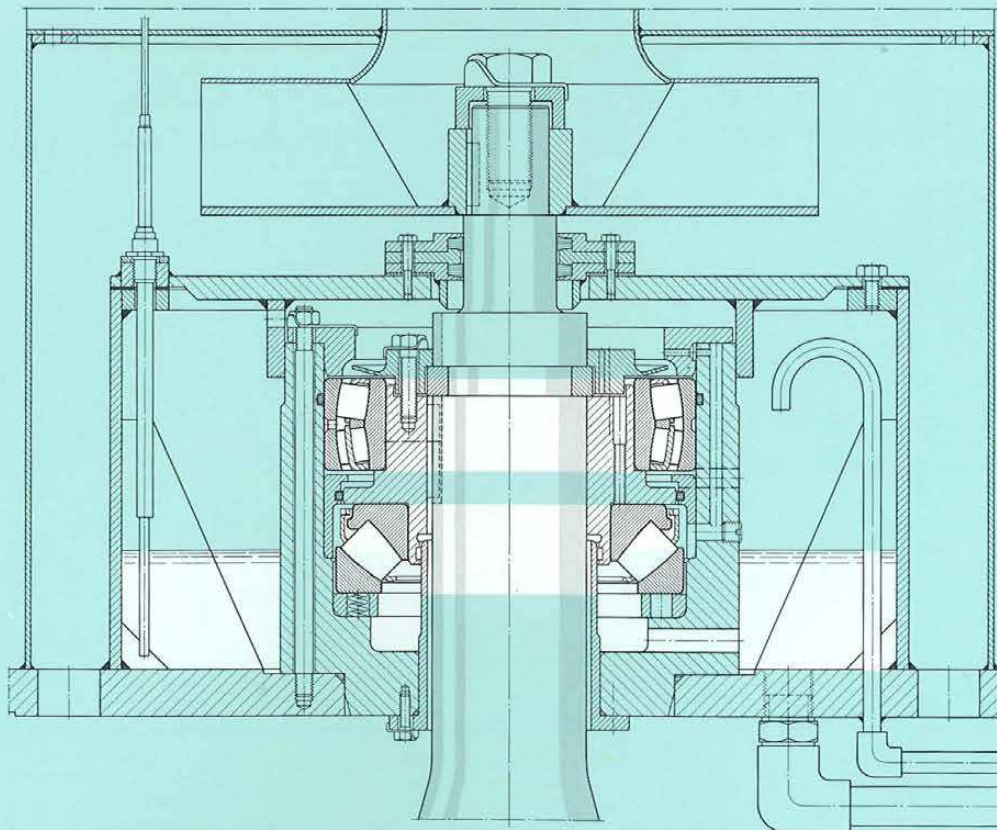
Tolerances

Shaft	Deep groove ball bearings	
	$d \leq 100 \text{ mm}$	k5
	$d > 100 \text{ mm}$	k6
	Cylindrical roller bearings	
	$d \leq 140 \text{ mm}$	m5
	$d > 140 \text{ mm}$	n6
	Spherical roller bearings	
Housing	$d \leq 140 \text{ mm}$	m6
	$d > 140 \text{ mm}$	n6
	Spherical roller thrust bearings	
	all diameters	j6
	Deep groove ball bearings (with O-ring to prevent creeping)	H7
	Cylindrical roller bearings	M7
	Spherical roller bearings (with O-ring to prevent creeping)	H7
	Spherical roller thrust bearings	clearance



Lubrication

Circulating oil lubrication is used for the bearings in the horizontal fan. Oil bath lubrication is preferred for the bearings in the vertical fan. The pumping action of the spherical roller thrust bearing is utilised to ensure lubrication of both bearings in this arrangement.



Appropriate housings

Y-bearing housings

Y-bearing	Appropriate housings				Complete units			
YEL 204	SY 504 M	P 47	FY 504 M	PF 47	SY 20 WM	P 20 WM	FY 20 WM	PF 20 WM
YEL 205	SY 505 M	P 52	FY 505 M	PF 52	SY 25 WM	P 25 WM	FY 25 WM	PF 25 WM
YEL 206	SY 506 M	P 62	FY 506 M	PF 62	SY 30 WM	P 30 WM	FY 30 WM	PF 30 WM
YEL 207	SY 507 M	P 72	FY 507 M	PF 72	SY 35 WM	P 35 WM	FY 35 WM	PF 35 WM
YEL 208	SY 508 M	P 80	FY 508 M	PF 80	SY 40 WM	P 40 WM	FY 40 WM	PF 40 WM
YEL 209	SY 509 M	P 85	FY 509 M	PF 85	SY 45 WM	P 45 WM	FY 45 WM	PF 45 WM
YEL 210	SY 510 M		FY 510 M	PF 90	SY 50 WM		FY 50 WM	PF 50 WM

SNA and SD plummer block housings

Bearings

Appropriate housing¹⁾

1205 KTN9/C3				SNA 505 TA, TC or TG
1206 KTN9/C3	2206 K/C3			SNA 506 TA, TC or TG
1207 KTN9/C3	2207 KTN9/C3	22207 CCK		SNA 507 TA, TC or TG
1208 KTN9/C3	2208 K/C3	22208 CCK	22208 CCK/C3	SNA 508 TA, TC or TG
1209 KTN9/C3	2209 K/C3	22209 CCK	22209 CCK/C3	SNA 509 TA, TC or TG
1210 KTN9/C3	2210 K/C3	22210 CCK	22210 CCK/C3	SNA 510 TA, TC or TG
1211 KTN9/C3	2211 K/C3	22211 CCK	22211 CCK/C3	SNA 511 TA, TC or TG
1212 K/C3	2212 K/C3	22212 CCK	22212 CCK/C3	SNA 512 TA, TC or TG
1213 KTN9/C3	2213 K/C3	22213 CCK	22213 CCK/C3	SNA 513 TA, TC or TG
1215 K/C3	2215 K/C3	22215 CCK	22215 CCK/C3	SNA 515 TA, TC or TG
1216 K/C3	2216 K/C3	22216 CCK	22216 CCK/C3	SNA 516 TA, TC or TG
1217 K/C3	2217 K/C3	22217 CCK/W33	22217 CCK/C3W33	SNA 517 TA, TC or TG
1218 K/C3	2218 K/C3	22218 CCK/W33	22218 CCK/C3W33	SNA 518 TA, TC or TG
1220 K/C3	2220 KM/C3	22220 CCK/W33	22220 CCK/C3W33	SNA 520 TA, TC or TG
1222 K/C3		22222 CCK/W33	22222 CCK/C3W33	SNA 522 TA, TC or TG
		22224 CCK/W33	22224 CCK/C3W33	SNA 524 TA, TC or TG
		22226 CCK/W33	22226 CCK/C3W33	SNA 526 TA, TC or TG
		22228 CCK/W33	22228 CCK/C3W33	SNA 528 TA, TC or TG
		22230 CCK/W33	22230 CCK/C3W33	SNA 530 TA, TC or TG
		22232 CCK/W33	22232 CCK/C3W33	SNA 532 TA, TC or TG
1306 K/C3				SNA 507 TA, TC or TG
1307 K/C3	2307 K/C3			SNA 508 TA, TC or TG
1308 K/C3	2308 K/C3	22308 CCK	22308 CCK/C3	SNA 510 TA, TC or TG
1309 K/C3	2309 K/C3	22309 CCK	22309 CCK/C3	SNA 511 TA, TC or TG
1310 K/C3	2310 K/C3	22310 CCK	22310 CCK/C3	SNA 512 TA, TC or TG
1311 K/C3	2311 K/C3	22311 CCK	22311 CCK/C3	SNA 513 TA, TC or TG
1312 K/C3	2312 K/C3	22312 CCK	22312 CCK/C3	SNA 515 TA, TC or TG
1313 K/C3	2313 K/C3	22313 CCK	22313 CCK/C3	SNA 516 TA, TC or TG
1315 K/C3	2315 K/C3	22315 CCK/W33	22315 CCK/C3W33	SNA 518 TA, TC or TG
1316 K/C3	2316 K/C3	22316 CCK/W33	22316 CCK/C3W33	SNA 519 TA, TC or TG
1317 K/C3	2317 KM/C3	22317 CCK/W33	22317 CCK/C3W33	SNA 520 TA, TC or TG
		22320 CCK/W33	22320 CCK/C3W33	SNA 524 TA, TC or TG
			23134 CCK/C3W33	SD 3134 TS
			23136 CCK/C3W33	SD 3136 TS
			23138 CCK/C3W33	SD 3138 TS
			23140 CCK/C3W33	SD 3140 TS
			23144 CCK/C3W33	SD 3144 TS
			23148 CCK/C3W33	SD 3148 TS
			23152 CCK/C3W33	SD 3152 TS
			23156 CCK/C3W33	SD 3156 TS
			23160 CACK/C3W33	SD 3160 TS
			23164 CACK/C3W33	SD 3164 TS

¹⁾ SNA housings offer a choice of three types of seal, namely type TA with two V-rings + washers, type TC with two felt seals and type TG with two double-lip seals. When ordering, the type of seal required is indicated by adding the type designation to the housing designation, e.g. SNA 505 TA.

Two-bearing housings

Bearing	Appropriate housing	Complete unit	Bearing	Appropriate housing	Complete unit
6206	PDN 206	PDNB 206			
6207	PDN 207	PDNB 207			
6208	PDN 208	PDNB 208			
6210	PDN 210	PDNB 210			
6211	PDN 211	PDNB 211			
6212	PDN 212	PDNB 212			
6214	PDN 214	PDNB 214			
6215	PDN 215	PDNB 215			
6216	PDN 216	PDNB 216	6216/C3	PDN 216	PDNB 216/C3
6218	PDN 218	PDNB 218	6218/C3	PDN 218	PDNB 218/C3
6220	PDN 220	PDNB 220	6220/C3	PDN 220	PDNB 220/C3
6222	PDN 222	PDNB 222	6222/C3	PDN 222	PDNB 222/C3
6224	PDN 224	PDNB 224	6224/C3	PDN 224	PDNB 224/C3
6305/C3	PDN 305	PDNB 305/C3	6305	PDN 305	PDNB 305
6306/C3	PDN 306	PDNB 306/C3	6306	PDN 306	PDNB 306
6307/C3	PDN 307	PDNB 307/C3	6307	PDN 307	PDNB 307
6308/C3	PDN 308	PDNB 308/C3	6308	PDN 308	PDNB 308
6309/C3	PDN 309	PDNB 309/C3	6309	PDN 309	PDNB 309
6310/C3	PDN 310	PDNB 310/C3	6310	PDN 310	PDNB 310
6311/C3	PDN 311	PDNB 311/C3	6311	PDN 311	PDNB 311
6312/C3	PDN 312	PDNB 312/C3	6312	PDN 312	PDNB 312
6313/C3	PDN 313	PDNB 313/C3	6313	PDN 313	PDNB 313
6314/C3	PDN 314	PDNB 314/C3	6314	PDN 314	PDNB 314
6315/C3	PDN 315	PDNB 315/C3			
6316/C3	PDN 316	PDNB 316/C3			
6317/C3	PDN 317	PDNB 317/C3			
6318/C3	PDN 318	PDNB 318/C3			
6319/C3	PDN 319	PDNB 319/C3			
6320/C3	PDN 320	PDNB 320/C3			
6322/C3	PDN 322	PDNB 322/C3			
6324/C3	PDN 324	PDNB 324/C3			

Bearings Locating side ¹⁾	Non-locating side	Appropriate housing	Complete unit
7214 BG	NU 2214 ECP/C3	PDP 214	PDPF 2214/C3
7216 BG	NU 2216 ECP/C3	PDP 216	PDPF 2216/C3
7218 BG	NU 2218/C3	PDP 218	PDPF 2218/C3
7220 BG	NU 2220/C3	PDP 220	PDPF 2220/C3
7222 BG	NU 2222/C3	PDP 222	PDPF 2222/C3
7224 BGM	NU 2224/C3	PDP 224	PDPF 2224/C3

¹⁾ Two bearings in back-to-back arrangement

Two-bearing housings

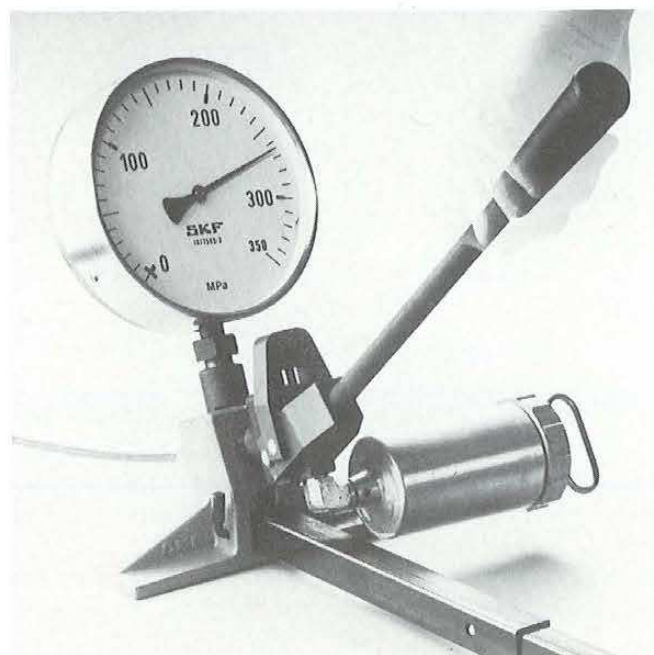
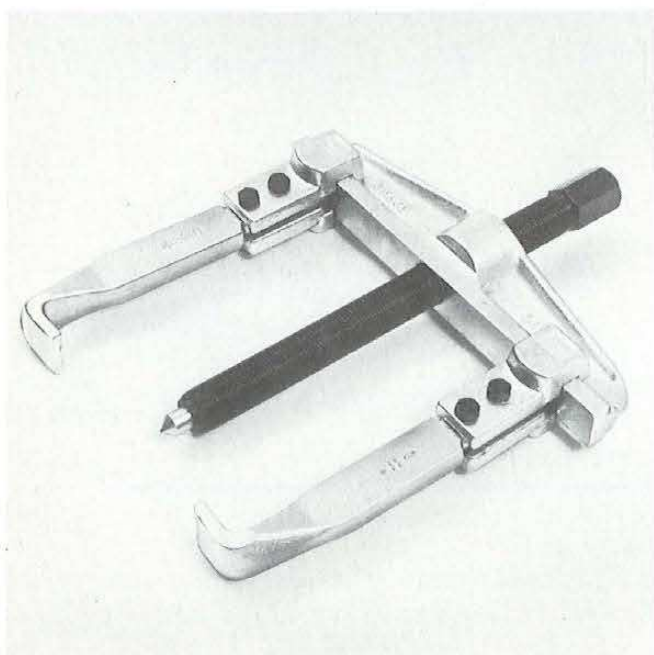
Bearings Locating side	Non-locating side	Appropriate housing	Complete unit
NU 315 ECP/C3 + 6315/C3	NU 315 ECP/C3	PDR 315	PDRJ 315/C3
NU 316 ECP/C3 + 6316/C3	NU 316 ECP/C3	PDR 316	PDRJ 316/C3
NU 317/C3 + 6317/C3	NU 317/C3	PDR 317	PDRJ 317/C3
NU 318/C3 + 6318/C3	NU 318/C3	PDR 318	PDRJ 318/C3
NU 319/C3 + 6319/C3	NU 319/C3	PDR 319	PDRJ 319/C3
NU 320/C3 + 6320/C3	NU 320/C3	PDR 320	PDRJ 320/C3
NU 322/C3 + 6322/C3	NU 322/C3	PDR 322	PDRJ 322/C3
NU 324/C3 + 6324/C3	NU 324/C3	PDR 324	PDRJ 324/C3

SOFN plummer block housings

Bearing	Appropriate housing	Bearing	Appropriate housing	Bearing	Appropriate housing
22217 CC/C3W33	SOFN 217			22217 CCK/C3W33	SOFN 517
22218 CC/C3W33	SOFN 218	22318 CC/C3W33	SOFN 318	22218 CCK/C3W33	SOFN 518
22220 CC/C3W33	SOFN 220	22320 CC/C3W33	SOFN 320	22220 CCK/C3W33	SOFN 520
22222 CC/C3W33	SOFN 222	22322 CC/C3W33	SOFN 322	22222 CCK/C3W33	SOFN 522
22224 CC/C3W33	SOFN 224	22324 CC/C3W33	SOFN 324	22224 CCK/C3W33	SOFN 524
22226 CC/C3W33	SOFN 226	22326 CC/C3W33	SOFN 326	22226 CCK/C3W33	SOFN 526
22228 CC/C3W33	SOFN 228			22228 CCK/C3W33	SOFN 528
22230 CC/C3W33	SOFN 230			22230 CCK/C3W33	SOFN 530
22232 CC/C3W33	SOFN 232			22232 CCK/C3W33	SOFN 532
22234 CC/C3W33	SOFN 234			22234 CCK/C3W33	SOFN 534
22236 CC/C3W33	SOFN 236			22236 CCK/C3W33	SOFN 536
22238 CC/C3W33	SOFN 238			22238 CCK/C3W33	SOFN 538
22240 CC/C3W33	SOFN 240			22240 CCK/C3W33	SOFN 540
22244 CC/C3W33	SOFN 244			22244 CCK/C3W33	SOFN 544

Mounting and dismounting

Reference should be made to the mounting and dismounting section of the SKF General Catalogue, but a more comprehensive treatment of the subject will be found in SKF publication Product Information 300 which also contains details of a variety of specially designed mounting and dismounting tools.



Relevant SKF publications

The publications listed below are only a few of many issued by SKF. If specific problems arise or if information not covered in these publications is needed, please contact your nearest SKF company or dealer.

- PI 102 SNA plummer blocks
- PI 106 Spherical roller thrust bearings of E design
- PI 111 Standard cylindrical roller bearings, ECP design
- PI 113 Two-bearing housings
- PI 300 Methods and tools for mounting and dismounting rolling bearings
- PI 302 Precautions when mounting rolling bearings
- WTM 15 355 SOFN plummer blocks

- 3200 General Catalogue
- 3289 Y-bearing units

Ball Bearing Journal

A quarterly review of rolling bearing engineering.





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